COMPUTER NETWORKS · CHAPTERS 1–5

1 · OVERVIEW & LAYERS

- 4-layer model: **Application → Transport → Internetwork (IP) → Link (LAN)**. Add **physical** = 5 layers. **OSI** inserts **Session + Presentation** between Transport and App.
- Each layer talks only to the layer above/below and adds a **header** (encapsulation). Typical headers: **Ethernet 14 B, IP 20 B, TCP 20 B**.
- **Data rate** = bits/s on a link. **Throughput** = effective rate incl. overhead. **Goodput** = useful app data (retransmits counted once). **Bandwidth** ≈ data rate.
- Rates: **kbps = 10^3 bits**, **Mbps = 10^6 bits**. File sizes: **kB = 2^10 B**, **MB = 2^20 B**.
- **Packet** = modest byte sequence + header. Packet switching: **Baran 1962** (survivability), **Davies 1964** coined "packet". Max payload: **Ethernet 1500 B**, **IP must support 576 B**, **ATM 48 B**, Token Ring 4 kB.
- **Datagram forwarding:** table of **(destination, next_hop)**; each switch does one hop; **stateless**, forwards each packet "in isolation" (**RFC 1122**).
- **Switch** = forwards by LAN/MAC address. **Router** = forwards by IP address. **Default entry** = 0.0.0.0/0 catch-all.
- **Virtual circuit** = the alternative: per-connection state; packets carry a connection ID instead of a destination.
- **Topology:** tree/acyclic = no redundancy. Loops give redundancy but allow **routing loops** (also "linear loop" between two neighbours). Ethernet forbids loops + never forwards back out the arrival port.
- **TTL:** 1-byte IP field, set by sender, **−1 per router**, dropped at 0 (usually starts 64). Limits loop damage.

1 · CONGESTION, DELAY, WINDOWS

- **Congestion:** in-rate > out-rate → queue; queue full → **drop**. Most Internet loss is congestion, not corruption.
- **Knee** = queue starts building (contention). **Cliff** = queue full, losses, throughput collapses.
- **Store-and-forward:** a router reads the whole packet before forwarding it.
- **Delay = bandwidth + propagation + store-and-forward + queuing.** Queuing is the only part engineering can reduce.
- **Bandwidth delay** = bits/rate per link. **Propagation** ≈ **200 m/µs** (⅔ c). **Store-and-forward** = sum of router bandwidth delays.
- **Sliding window:** keeps W packets in flight; ideal W sends one window per **RTT**. **TCP Reno** = grow until loss → **halve** → sawtooth.
- **Head-of-line blocking:** one lost TCP packet stalls everything behind it (bad for real-time; UDP avoids it).

1 · ADDRESSES, IP, TRANSPORT

- **MAC** = 48-bit, burned in; **first 3 B = OUI** (vendor); no location meaning. Broadcast = all 1s; unicast = one; multicast = group.
- **IPv4** = 32-bit, **network prefix + host**. Legacy classes: **A /8** (0…), **B /16** (10…), **C /24** (110…), **D** = multicast (1110).
- Modern = **CIDR** a.b.c.d/n. Network byte order = **big-endian** (**RFC 1700**). IP is **best-effort, connectionless**; no ACKs/retransmits; supports **fragmentation**.
- Prefixes make tables scale: ~**800k** prefixes vs ~10^9 hosts. IANA ran out of /8 in **2011**; ARIN in **2015**.
- **DNS:** hierarchical, distributed, cached; name → IP.
- **TCP:** reliability (seq #, ACKs, timeout/retransmit, reorder), connection-oriented, stream-oriented, **ports**, throughput management, sliding windows.
- **Socket = (host, port)**. Ports: **80 = web**, **25 = email**. A connection is identified by both endpoints' sockets.
- **UDP:** ports + checksum only; no setup, no loss detection, no retransmit. Used for real-time; **RTP** sits above it.
- **CDN:** replicate content at edge servers (PoPs); map users by (1) different IPs + DNS, (2) same IP + **anycast**, (3) HTTP redirect.
- **Firewall** blocks risky inbound connections; **NAT** shares one public IP and adds firewall protection; **buffer overflow** is a classic total-compromise attack.
- Tools: **ping** (ICMP), **ifconfig/ipconfig/ip**, **nslookup/dig/host**, **traceroute/tracert**, **route/netstat -r**, **netcat/nc**, **Wireshark/tcpdump**.
- Orgs: **IETF** (under IAB/ISOC) publishes **RFCs**; **RFC 2119** MUST vs SHOULD; **Postel**: "conservative in what you send, liberal in what you accept."
- History: ARPANET **1969** (DARPA), NSFNet **1986**, ISOC **1992**. OSI (ISO, **1977**) = 7 layers. IETF = "rough consensus and running code"; **2 interoperable implementations** rule.
- **Wasp waist:** many protocols above and below, a single IP layer. **Berkeley Unix** = de facto reference; TCP Tahoe/Reno named after **4.3BSD** releases.

2 · ETHERNET

- **Metcalfe & Boggs 1976**; 10 Mbps → 100 → 1 Gbps → 10 G. **CSMA/CD** = Carrier Sense, Multiple Access, Collision Detect.
- Evolution: coax bus → **repeater** (per-bit) → **hub** (multiport repeater, *propagates* collisions) → **switch/bridge** (buffers whole packet, *does not* propagate collisions; isolates **collision domains**).
- Frame: **dest 48b | src 48b | type 16b | data ≤1500 | CRC-32**. Type: **0x0800 IP, 0x0806 ARP, 0x8137 IPX**. 802.3 replaced type with length. Preamble syncs.
- IEEE: **802.3 = Ethernet, 802.11 = Wi-Fi, 802.16 = WiMAX**.
- Address bits: **LSB of first byte = multicast**; **second bit = universal/local**. NIC accepts if own addr, broadcast, subscribed multicast, or **promiscuous**.
- **Slot time 512 bits = 64 B = 51.2 µs** (10 Mbps); max diameter **232 bits**; **min frame 64 B (46 data)**; **jam 48 bits**; **96-bit (9.6 µs) gap**; max **1024 hosts**.
- **Backoff:** after attempt N (1–10) pick **k < 2^N** slots; N 11–15 use k<1024; **give up at N=16**.
- **Capture effect:** backoff is unfair; a repeated winner gets better odds. Fixed by BLAM / full-duplex.
- PHY: 10M **Manchester** (20 M electronics); 100BASE-TX **4B/5B**, Cat-5, 100 m; **full-duplex = no collisions**; 1000BASE-T **PAM-5**, 4 pairs × 250 Mbps, 4096-bit slot; 10G abandons collisions.
- **1-persistent** CSMA (transmit as soon as idle) → guaranteed initial collision; handled by backoff. **ALOHA** max throughput **1/2e ≈ 18%**; Ethernet ~**64–80%**.

2 · SWITCHING

- **Learning algorithm:** on arrival record **(source, arrival port)**; forward a known dest out its port; **flood** an unknown dest out all other ports; if dest = arrival port, drop. Entries expire (~5 min).
- Tables ~**10^4–10^5** entries; keep **≤1000 hosts/LAN**. Flooding = "unknown unicast flooding"; this fallback is what lets switches learn with no protocol.
- **MAC flooding attack:** fake source addresses overflow tables → flooding → eavesdropping. Defense: **port security** (limit entries per port).
- Switches give privacy (traffic only where needed) and kill collisions, but **broadcast traffic still goes everywhere**.

3 · ADVANCED ETHERNET

- **Spanning Tree (Perlman 1985):** elect **root = smallest switch ID**; exchange **BPDUs** ("Hello") to **01:80:c2:00:00:00** carrying (switch ID, believed root, cost).
- Enable: (1) port toward root, (2) ports others use toward root, (3) shared segment: closest-to-root switch (tie → smaller ID), (4) stub/host ports. Prune the rest.
- Disabled links idle even if shorter; traffic goes via/near the root; **~30 s** convergence. Rogue root → **Root Guard**.
- **VLAN:** logically separate groups; **802.1Q tag** = 32-bit (12-bit VLAN ID + 3-bit priority), EtherType **0x8100**, inserted after source. Inter-VLAN needs a **router**. ~**256 hosts (/24)** per VLAN.
- **Switch hardware:** shared-memory (CPU copies in/out, needs 2× rate), **crossbar** (N×N parallel paths; input + output queues), **CAM/TCAM** (single-load lookup; TCAM masks = IP prefixes, longest match, shorter prefixes first).
- **TRILL/SPB:** keep **all links active**, route shortest paths. **RBridges** run **IS-IS**; each Legacy Ethernet elects a **DRB**; 2-byte **nicknames**; TRILL header has **hopcount**; recovery **<0.1 s**; **ECMP**. SPB = 802.1aq, no hopcount.
- **SDN:** central **controller** programs **flow tables** = match fields + actions (drop, forward, flood, send to controller, modify, goto table). **Priority**; table-miss = priority 0. Counters + last_used. NO_FLOOD ports. Learning via (dst,src) pairs (N^2) or 2 tables (dest, then src). Uses: L3 switch, load balancer, firewall, KV store.

4 · WIRELESS / WI-FI

- Radio problems: easy eavesdropping (encrypt), **cannot detect collisions** (~−60 dB while transmitting), **hidden nodes**, collisions are **local** (at the receiver), shared spectrum cost, **multipath**/fading (ripples ~½ wavelength), power.
- **Shannon–Hartley:** max rate / channel width = **log2(1+SNR)**. **OFDM** = many narrow subchannels; **OFDMA** = multiple users (802.11ax).
- Wi-Fi = **802.11 a/b/g/n/ac/ax** = Wi-Fi **4/5/6**. Rates/bands: **a 54 Mbps 5 GHz; b 11 Mbps 2.4; g 54 Mbps 2.4; n 65–150 2.4/5; ac 78–867 5 GHz; ax ≤1200 2.4/5/6**.
- 2.4 GHz ISM: use channels **1/6/11**; 5 GHz has many more channels.
- **CSMA/CA** (avoid, not detect): unicast **link-layer ACK** (no seq #) after **SIFS**; no ACK → assume collision → retransmit.
- Timings (802.11b/g): **slot 20 µs, DIFS/IFS 50 µs, SIFS 10 µs**. Backoff starts **k<32**, doubles to 1024, ~7 tries; clock suspends during others' traffic.
- **RTS/CTS:** optional for big packets; the CTS tells hidden nodes how long to stay quiet. **Fragmentation** for high error rates.
- **Rate scaling:** 54/48/36/24/18/12/9/6 Mbps; **ARF** (down after 2 fails, up after 10); preambles + ACKs at min rate.
- **MIMO:** multiple antennas → multiple **spatial streams**; multipath makes paths independent; SIMO/MISO/beamforming; 802.11n ≤4 (600 Mbps), 802.11ac ≤8 (~7 Gbps); notation **A×B×C**.
- **AP:** infrastructure vs **ad hoc**; **SSID**; **beacon every 100 ms**; join = scan/probe → open-auth+association → real auth → **DHCP**. **MAC randomization** (per-connection/per-network).
- **Roaming:** same SSID = **ESS**; **reassociation** request; **802.11r** fast handoff. **Mesh 802.11s:** mesh BSS, **HWMP**, root mesh station; EasyMesh Controller+Agents.
- **Security: WEP** (RC4, broken) → **WPA** (TKIP) → **WPA2** (AES/**CCMP**, 802.11i) → **WPA3** (**SAE**, forward secrecy).
- **WPA2-Personal** = shared **PSK**; **Enterprise** = **802.1X/EAP/RADIUS**, roles supplicant/authenticator/auth server, **EAPOL**.
- **Four-way handshake:** ANonce + SNonce → **PTK** = hash(master, nonces, MACs); **MIC** proves the key; replay counter; optional **GTK**. **KRACK** = replay msg 3 → nonce reinstall → keystream reuse; fix = ignore replays. **802.11w** protects management frames.
- **PCF** (polling, **CFP**, PIFS) vs **DCF**; rarely used. **MANET** = mobile ad hoc; needs routing; no broadcast/multicast by default.
- **WiMAX 802.16 / LTE (E-UTRA, eNB):** "4G" = **IMT-Advanced**; licensed spectrum; coverage 1–10 km. Base station **schedules uplink** (no collisions); slots **LTE 1 ms, WiMAX 2 ms**; **ranging** = distance → timing + power; **network entry is the only collision time** (LTE **RACH**, 64 orthogonal preambles → **RNTI**).
- **Fixed wireless:** terrestrial (line-of-sight, dishes) and **satellite** (geosynchronous **35,786 km**, RTT ~**500 ms** + queuing ≈1000 ms; data caps).

5 · OTHER LANs

- **VPN: tunneling**; software virtual interface (e.g. tun0) gets an internal IP; encapsulated (± encrypted) over the Internet; keep a **host-specific route** to the VPN server; often default-route through the tunnel. **ECN re-marking: RFC 6040**.
- **Carrier Ethernet:** leased point-to-point that looks like Ethernet at both ends; the middle can be anything; full-duplex, collision-free, arbitrary length.
- **Token Ring:** a **token** circulates; transmit only when holding it → **collision-free**, **fair** (N stations get 1/N). FDDI/IBM; monitor lost/duplicate token. **Token Bus** = Ethernet + virtual ring (failed).
- **Virtual circuit:** set up a connection first; packets carry a **VCI**; **VCI unique per link** and **rewritten at each switch**; table keyed **(VCI, port) → (VCI, port)**. Advantages: **QoS guarantees**, small headers, tiny voice packets. Datagram advantages: less state, crash-resilient, easy reroute, hard to bill. RSVP/MPLS.
- **ATM: 53 B cell = 48 data + 5 header**; small cells → low voice **fill time** and queuing; fixed size; **no cell reordering**, no broadcast. Header: VPI+VCI ≤28 bits, 3 type bits, **CLP**, 8-bit HEC.
- **AAL1/2** voice. **AAL3/4:** 44-B chunks, 4-B AAL header/trailer (type, seq, MID), **CRC-10** → **>20% overhead**. **AAL5:** 48-B chunks, **CRC-32** on CS-PDU, **10% overhead**, detects lost cells. **Moral: one checksum over the whole message beats many short ones.**

FORMULAS & PROCEDURES

- **Forwarding table:** next hop = neighbour on a **shortest path** (BFS); direct/local if dest is on this switch; ties → smaller ID. Store only the **first** hop.
- **Learning switch:** learn source+port; forward if dest known; **flood** if unknown; drop if dest port = arrival port.
- **Spanning tree:** root = min ID; compute the shortest-path tree; **prune** every edge not in it.
- **Backoff:** attempt N → random **k < 2^N** slots (cap 1024); give up at 16.
- **Longest-prefix match:** most specific matching prefix wins; **/0** only if nothing else matches.
- **VCI:** look up **(VCI_in, port_in) → rewrite VCI_out, send out port_out**.
- **ATM cells:** AAL5 = **ceil(P/48)**; AAL3/4 = **ceil((P+8)/44)** . Overhead 5/53 ≈ 9.4% (AAL5) or 9/53 ≈ 17% (AAL3/4).
- **Token Ring wait ≈ (N/2) × token_time**, token_time = token_bytes×8/rate.
- **Contention** = slots × 20 µs; tx = bytes×8/rate (Mbps → µs).
- **Delay** = (1+R)×(bytes×8/rate) + km/200.
- **Ranging guard** = 2×(v×T)/300 µs.
- **Sliding-window throughput** = W×size×8/RTT.

NUMBERS TO RECOGNIZE

1500 B Ethernet · **576 B** min IP · **64 B** min frame / **46** data · **48 B** ATM data (+5) · **51.2 µs** slot · **232-bit** diameter · **96-bit** gap · **1024** hosts · **N=16** give up · **20/50/10 µs** Wi-Fi · **k<32** start · **48-bit** MAC · **32-bit** IPv4 · **35,786 km** satellite · **200 m/µs** propagation · **30 s** STP · **0.1 s** TRILL · **100 m** Cat-5 · **4096-bit** gigabit slot · **10^4–10^5** switch entries.